Intrinsic Angular Momentum of Nucleons and Leptons: Constituent-Quark Zitterbewegung and a Postulated Superluminal Torsion Background

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Abstract

The spin of the nucleon is, in standard QCD, a sum of quark spin, gluon spin and orbital contributions, with the quark-spin share measured at $\Delta\Sigma \approx 0.30 \pm 0.05$. This paper keeps that decomposition and adds a structured picture with explicit (mostly null-test) predictions of the "trembling" (rapid micro-motion) of fundamental fermions. We adopt, as a working identification [H], that the trembling is the Dirac *Zitterbewegung* (ZB) of the relevant elementary degrees of freedom: for the electron, which has no constituents, it is the ZB of the electron itself, with $\omega_Z = 2m_e c^2/\hbar = 1.553 \times 10^{21}$ rad/s (at $p = 0$); for the proton and neutron it is the ZB of the constituent quarks (~ 313 MeV), with $\omega_Z = 9.51 \times 10^{23}$ rad/s and amplitude $r_Z = 0.315$ fm (at $p = 0$, an order-of-magnitude value for a confined quark). The trembling is therefore set by the mass (and, for a particle with momentum, by $E(p)$) and is *not produced by the torsion particle*; any τ_T effect on it is at most a small perturbation, not evaluated here. Within the TauTimp framework the Entanglement–Torsion quantum τ_T is retained, but only as a hypothetical background that can at most produce a small, testable perturbation (spin precession or relaxation), to be bounded by NMR and ESR linewidths and the electron $g - 2$. We postulate that τ_T propagates at $V = 10^9 c \approx 3.0 \times 10^{17}$ m/s. This is an assumption, not a result: we compute its direct consequences (transit times, retardation, momentum ambiguity, frame anisotropy) and state openly its conflicts with Lorentz invariance, causality, and no-signalling. No experimental confirmation of τ_T or of $V = 10^9 c$ is claimed. The "fifth fundamental force" is presented as a hypothesis with explicit predictions. Corrections to numerical values of the May 2026 version are collected in Appendix B.

Keywords: spin decomposition, Zitterbewegung, constituent quarks, torsion, superluminal postulate, TauTimp

1. Introduction and Status of Claims

Spin is among the most precisely measured properties of elementary particles, but its interpretation is still debated. Dirac's equation yields spin-$1/2$ as a consequence of Lorentz covariance [4]; the "spin crisis" showed that quark helicities carry only a fraction of the proton spin [3, 5, 1]. The present work asks two questions: (i) what is the "trembling" of fermions, and which objects produce it; (ii) what can be said, without overreach, about a hypothetical torsion background propagating at $V = 10^9 c$.

Relative to the May 2026 version [14], this revision (a) assigns the trembling to Dirac ZB of the elementary degrees of freedom (constituent quarks for nucleons; the electron itself for the electron), not to τ_T; (b) treats $V = 10^9 c$ as an explicit postulate, replacing the earlier statement that τ_T moves at c; (c) corrects numerical values that were not reproducible from the printed formulas (Appendix B); (d) presents the "fifth force" as a hypothesis (Section 6).

Key claims carry one of the following labels.

Label	Meaning
[E]	Established: standard physics, experimentally supported
[M]	Mathematical identity: follows exactly from the stated formulas
[I]	Interpretation: model-dependent reading of [E]/[M]
[T]	Taxonomy: TauTimp bookkeeping, not Standard Model physics
[P]	Postulate: assumed, not derived
[H]	Hypothesis: testable proposal, no evidence yet

2. Nucleon Spin in QCD 2.1 Decomposition [E]

The total angular momentum of a nucleon is decomposed as

$$\vec{J} = \vec{S}_q + \vec{L}_q + \vec{S}_g + \vec{L}_g,$$

with quark spin, quark orbital, gluon spin and gluon orbital terms. In the sum-rule form for a spin-$1/2$ nucleon,

$$1/2 = 1/2\ \Delta\Sigma + L_q + \Delta G + L_g, \qquad \Delta\Sigma \equiv 2\langle S_q \rangle \approx 0.30 \pm 0.05,$$

so roughly 70% of the nucleon spin is carried by gluon helicity and orbital motion of quarks and gluons [3, 5, 1]. The separation of L_q and L_g depends on gauge and scheme choices, and $\Delta\Sigma$ is defined at a stated renormalization scale; see the review in [1].

2.2 Informational-quark taxonomy [T]

The TauTimp framework [13] uses six "informational quarks": Mass B, Magnetic field D, Electric charge S, Gravity C, Light U, Time T. These labels are unrelated to QCD quarks and flavours and are **not** part of the Standard Model. The framework assigns:

Particle	Mass (B)	Mag. Field (D)	El. Charge (S)	Gravity (C)	Light (U)	Time (T)
Proton	+1836	+2.793	+1	0	0	0

Particle	Mass (B)	Mag. Field (D)	El. Charge (S)	Gravity (C)	Light (U)	Time (T)
Neutron	$+1838$	-1.913	0	-1	0	0
Electron	$+1$	$+1838$	-1	0	0	0

The D entries for the proton and neutron coincide with their measured magnetic moments in nuclear magnetons, $+2.793\ \mu_N$ and $-1.913\ \mu_N$ **[E]**. The table is a mnemonic encoding and carries no dynamical content in this paper.

2.3 The electron is not a composite [E]

The May 2026 version attributed the spin of the proton, neutron *and electron* to quark–gluon dynamics. This is withdrawn for the electron. In the Standard Model the electron is an elementary lepton with no quark or gluon content; its spin $\hbar/2$ is a property of the elementary Dirac field. Only the nucleon spin has a quark–gluon decomposition.

3. Trembling as Dirac Zitterbewegung 3.1 The Dirac result

The Dirac equation [4] **[E]** describes a free spin-$1/2$ particle of mass m. Interference between its positive- and negative-energy solutions makes the position oscillate (Schrödinger 1930 [2]). The velocity operator has eigenvalues $\pm c$, and, for a particle at rest ($p = 0$), **[M]**

$$\omega_Z = \frac{2mc^2}{\hbar}, \qquad r_Z = \frac{\hbar}{2mc} = 1/2\ \lambda_C, \qquad \omega_Z\, r_Z = c,$$

where $\lambda_C = \hbar/mc$ is the reduced Compton wavelength. For $p \neq 0$ the frequency generalizes to $\omega_Z = 2E/\hbar$ with $E = \sqrt{p^2c^2 + m^2c^4}$ (Section 3.5). Identifying r_Z with the radius of the motion, in the circular-motion reading of Hestenes [6] **[I]**, the associated angular momentum is

$$L = m\, c\, r_Z = \frac{\hbar}{2},$$

exactly at $p = 0$, for any mass. The loop magnetic moment in this picture corresponds to $g = 1$; $g = 2$ requires the full Dirac current. ZB has **not been observed for free electrons**; it has been reproduced only in a quantum simulation with trapped ions [7], an analogue and not a measurement on a free electron.

3.2 Who trembles: elementary degrees of freedom

Electron. The electron is elementary, so its trembling is its *own* ZB. No quark enters.

Proton and neutron. We adopt, as a working identification **[H]**, that the trembling of a nucleon is the Dirac-type ZB of its three **constituent quarks**, each with an effective mass $m_q \approx 313$ MeV (a typical constituent-quark-model value, $\approx m_p/3$). Two limitations are explicit. First, quarks are confined: a free $+E/-E$ wave packet, which is what the ZB derivation of Section 3.1 requires, does not exist as such in a confining potential, and the free Dirac equation applies only formally. Second, the constituent mass is an effective quasi-particle mass (it includes gluon and sea-quark dressing), not the mass of an isolated quark. The identification is useful because the resulting length scale is hadronic, and it is weakly constrained (Section 6.3, item 3).

Table 1 gives values computed from the formulas of Section 3.1 with CODATA constants [11] ($\hbar = 1.054571817 \times 10^{-34}$ J s, $c = 2.99792458 \times 10^{8}$ m/s). Period is $2\pi/\omega_Z$; $D = c\, r_Z$.

Object	mc^2	ω_Z (rad/s)	r_Z	Period (s)	$D = \hbar/2m$ (m²/s)
Electron (elementary)	0.511 MeV	1.553×10^{21}	193.1 fm	4.05×10^{-21}	5.79×10^{-5}
Constituent quark	313 MeV	9.51×10^{23}	0.315 fm	6.6×10^{-24}	9.45×10^{-8}
Proton as a whole (formal)	938.27 MeV	2.85×10^{24}	0.105 fm	2.2×10^{-24}	—
u quark, current mass (formal)	2.16 MeV	—	45.7 fm	—	—

Table 1. Zitterbewegung parameters at $p = 0$ (free particle at rest). The last two rows are included only as a check on which mass is meaningful.

Which mass is meaningful. For the current u-quark mass, $r_Z = 45.7$ fm, about 50 times the proton radius ($r_p = 0.8409$ fm [12]); this is inconsistent with a confined quark. The constituent value, $r_Z = 0.315$ fm, is comparable to the proton radius, and the whole-proton value (0.105 fm) is the formal ZB of a point Dirac particle of proton mass. Hence the constituent-quark scale is the physically sensible one for nucleon trembling.

3.3 Spin and the QCD decomposition [I]

If each constituent quark carries $\hbar/2$ from its own ZB loop ($m_q c\, r_Z = \hbar/2$), the simple quark model combines three such spins into $1/2 + 1/2 - 1/2 = 1/2$. The measured $\Delta\Sigma \approx 0.30$ could then be read as a partial depolarization of the loops through gluon exchange **[H]**, with the remainder in L_q, ΔG and L_g. This reading is an interpretation and a hypothesis: ZB at the constituent scale and $\Delta\Sigma$ at the partonic scale are different effective descriptions, and this paper does not derive $\Delta\Sigma$.

3.4 Diffusion coefficient identity [M]

For ZB, $c\, r_Z = \hbar/2m$, which equals the diffusion coefficient in Nelson's stochastic mechanics [8], where Brownian motion with $D = \hbar/2m$ yields the Schrödinger equation. The *identity* is exact; the statement that ZB behaves as Brownian noise is a **hypothesis [H]** and is not used below.

3.5 Dependence on mass and momentum; consequence for the torsion particle

For a particle at rest ($p = 0$) ω_Z, r_Z and D are fixed by the mass. For $p \neq 0$ the frequency is $\omega_Z = 2E/\hbar$, $E = \sqrt{p^2c^2 + m^2c^4}$, so it is not fixed by the mass alone. Two cases:

- *Atomic electron.* With $p \sim \alpha m_e c$, $E \approx m_e c^2(1 + \alpha^2/2)$ and the correction is $\alpha^2/2 \approx 2.7 \times 10^{-5}$: negligible.
- *Confined constituent quark.* With $R = 0.8409$ fm [12] and $\hbar c = 197.327$ MeV fm, two confinement estimates of the momentum give: $\Delta p \geq \hbar/2R$, i.e. $pc \approx 117$ MeV, $E \approx 334$ MeV; and $p \approx \hbar/R$, i.e. $pc \approx 235$ MeV, $E \approx 391$ MeV. Accordingly ω_Z is larger by between $\approx 7\%$ and $\approx 25\%$, depending on the confinement estimate; the free-Dirac formula is itself only formal for a confined quark. This is an order-of-magnitude correction; it is **not used** in Table 1.

The role of τ_T is reduced to that of a possible small perturbation on top of ZB (Section 5). The ZB frequency itself is fixed by E, as above. Whether a passing τ_T could affect the ZB phase is discussed in Section 4.3(a), under the postulate P1 of Section 4.2.

4. The Torsion Particle and the Postulate $V = 10^9 c$

4.1 Taxonomy [T] and production [P]

Particle	Mass (B)	Mag. Field (D)	El. Charge (S)	Gravity (C)	Light (U)	Time (T)
Entanglement –Torsion (τ_T)	0	-1.913	0	0	$+1$	-1

These are bookkeeping labels of the framework [13]. They do not by themselves derive any speed or dispersion relation. In particular, the earlier inference "zero rest mass, therefore $V = c$" is withdrawn: the kinematics of τ_T is set by the postulate below.

Postulate P2 (stellar production) [P]. Stellar production of τ_T is a postulate of the framework, without a verified mechanism. The framework's notation for a model reaction is given, with an explicit statement of its problems, in Appendix A.

4.2 Postulate P1 [P]

Postulate P1. The torsion particle τ_T propagates, in a preferred reference frame $\mathcal{S}_0$, with constant speed

$$V = 10^9\, c \approx 2.998 \times 10^{17} \text{ m/s}.$$

We postulate P1. It is not derived from the taxonomy, not inferred from data, and **no experiment confirms it, or the existence of** τ_T. Its value (10^9) is a free parameter of the framework. The remainder of this section computes what follows if P1 holds, and what it conflicts with.

4.3 Direct consequences of P1

All values below are computed from V and the constants of Section 3.2 unless noted.

Quantity	At c	At $V = 10^9 c$
Speed	2.998×10^8 m/s	2.998×10^{17} m/s
Time to cross a proton, $2r_p/v$ ($r_p = 0.8409$ fm)	5.6×10^{-24} s	5.6×10^{-33} s
Same, relative to the constituent-quark ZB period (6.6×10^{-24} s)	≈ 0.85	$\approx 8.5 \times 10^{-10}$
Time to travel 1 Mpc	$\approx 3.3 \times 10^6$ yr	1.03×10^5 s (≈ 1.2 days)
Time to travel the Hubble radius (≈ 4.3 Gpc)	—	≈ 14 yr
Number density for a given flux, $\rho_\tau = \Phi_0/v$	Φ_0/c	$10^{-9}\,\Phi_0/c$
Momentum of a quantum of energy E_m	$p = E_m/c = 1.92 \times 10^{-27}$ kg m/s	see (d)

(Here $E_m = 5.751344 \times 10^{-19}$ J $= 3.59$ eV, $\nu = 8.68 \times 10^{14}$ Hz, is the framework's energy parameter.)

(a) Transit time versus ZB period. Under P1, and assuming an interaction range of order r_p, a passing τ_T acts impulsively on the nucleon (transit $\approx 10^{-9}$ of the ZB period). An impulsive action has a broad spectrum and may in principle excite ZB or shift its phase; this effect is not evaluated here. The ZB frequency itself is fixed by E (Section 3.5), not by τ_T; the only constraints on τ_T–ZB coupling used in this paper are the NMR/ESR and $g - 2$ bounds of Section 5.

(b) Retardation. A burst emitted at distance d would reach us after d/V, i.e. a lead of $\approx d/c$ over light. For $d = 1$ Mpc: 1.03×10^5 s by τ_T against ≈ 3.3 Myr by light (1.03×10^{14} s $= 1.03 \times 10^5$ s $\times\, 10^9$). The "isotropic and constant flux" of the earlier version then refers to an almost instantaneous snapshot of the whole observable stellar population, not to light-delayed sources.

(c) Flux and density. The earlier relation $\rho_\tau = \Phi_0/c$ becomes $\rho_\tau = \Phi_0/V$.

(d) The momentum is not determined by P1. The postulate gives a speed but no dispersion relation. Two natural choices differ by 10^{18}:

Assumption	Momentum of a E_m quantum
$p = E/V$ (photon-like relation with $c \to V$)	1.92×10^{-36} kg m/s

Assumption	Momentum of a E_m quantum
$p = EV/c^2$ (standard relation $v = pc^2/E$ for relativistic-type kinematics, as in tachyon models [9])	1.92×10^{-18} kg m/s

The second value is $E_m V/c^2 = 10^9\, E_m/c$. This paper does not select one; any prediction involving τ_T recoil depends on that choice.

(e) Anisotropy seen from a moving frame. If V holds in the rest frame $\mathcal{S}_0$, a frame moving at speed u relative to $\mathcal{S}_0$ sees a torsion flux whose anisotropy is of order u/V. Taking, as an illustration, $u = 3.7 \times 10^5$ m/s (the order of the solar-system velocity relative to the cosmic microwave background, assumed here to define $\mathcal{S}_0$, which is itself an extra assumption): $u/V \approx (3.7 \times 10^5 \text{ m/s})/(3.0 \times 10^{17} \text{ m/s}) \approx 1 \times 10^{-12}$.

4.4 Tensions with relativity (declared, not resolved)

1. **Lorentz invariance.** P1 singles out a frame $\mathcal{S}_0$ in which V takes its value. Matter and light obey Lorentz invariance to high precision; τ_T would break it in its own sector. Any coupling of τ_T to ordinary matter (such as $\mathcal{H}_\tau$ in Section 5.1) would bring the experimental bounds on Lorentz violation into play; they are not evaluated here.
2. **Causality.** If a signal travels at speed V in $\mathcal{S}_0$, the Lorentz transformation to a frame moving at speed u along the propagation direction gives $\Delta t' = \gamma\, \Delta t\, (1 - uV/c^2)$. This is negative when $u > c^2/V = c/10^9 \approx 0.30$ m/s. For such observers, in a Lorentz-covariant description, the signal arrives before it is emitted, and closed causal loops become possible if signals can be sent controllably (the standard argument for tachyons [9]). Ordinary laboratory motion exceeds 0.30 m/s. The postulate can avoid this only by asserting that (i) a preferred frame with absolute simultaneity exists and the Lorentz transformation does not apply to τ_T signals, and (ii) τ_T cannot be used as a controllable signal carrier. Both are additional assumptions, not results.
3. **No-signalling.** The theorem forbidding signalling by entanglement correlations does not protect against a real particle carrying energy (E_m) faster than light. If τ_T could be emitted and detected on demand, it would be a superluminal communication channel. Consistency requires that emission is uncontrollable (stellar) and detection is through the averaged background only; this is an assumption that must be stated with the theory, not an outcome of it.
4. **Energy loss.** A superluminal particle that couples to the electromagnetic field would be expected to lose energy through vacuum Cherenkov-type radiation. Whether τ_T avoids this (it has no charge but a magnetic-moment label) is not analyzed here.
5. **Flux isotropy.** Even with V large, the claim of isotropic, constant flux requires a calculation that includes the Sun, the Galaxy and the Local Group (an Olbers-type argument); it is asserted, not shown, in this paper.

Summary. P1 is logically admissible only within a preferred-frame theory in which τ_T is non-controllable. It is in tension with special relativity as usually formulated, and no data supports it.

5. Torsion as a Small Perturbation 5.1 Coupling and the isotropy problem [H]

The framework couples a torsion field $\vec{B}_\tau$ to the magnetic moment,

$$\mathcal{H}_\tau = -\vec{\mu} \cdot \vec{B}_\tau.$$

A coherent precession of a spin about a non-zero mean field has angular frequency

$$\omega_\tau = \frac{2\mu B_\tau}{\hbar}.$$

If the flux is truly isotropic, the *mean* vector field vanishes, $\langle \vec{B}_\tau \rangle = 0$, so there is no first-order coherent precession. The remaining effect is stochastic: a fluctuating isotropic field produces line broadening and relaxation (at second order in B_τ), not a rotation. A true precession requires a net direction, i.e. a departure from isotropy. The bound below applies to the mean component; relaxation limits are obtained from NMR/ESR linewidths.

5.2 Required field strengths

The earlier version used the formula $\omega = \sqrt{5Be\hbar/(mr^2)}$ for a "precession" frequency. Recomputed from that formula with the earlier inputs, $\omega_e = 6.67 \times 10^{10}$ s^{-1} and $\omega_p = 5.04 \times 10^9$ s^{-1} (the printed values were different; see Appendix B). A precession at these frequencies requires $B_\tau = \hbar\omega/2\mu$, with $\mu_B = 9.274 \times 10^{-24}$ J/T and $\mu_p = 1.411 \times 10^{-26}$ J/T [11]:

Particle	Frequency	B_τ required
Electron	6.67×10^{10} s^{-1}	≈ 0.38 T
Proton	5.04×10^9 s^{-1}	≈ 19 T

Such backgrounds are excluded: nuclear magnetic resonance works in the Earth's field (5×10^{-5} T), electron and nuclear resonance lines would be broadened beyond recognition, and the electron $g - 2$ constrains anomalous couplings of the electron further. Hence, **if τ_T exists, B_τ lies orders of magnitude below these values**. As a scale only: the proton precession frequency in the Earth's field is $2\mu_p B/\hbar \approx 1.3 \times 10^4$ rad/s, so any torsion-induced mean precession should be small compared to what can be resolved in such experiments. The numerical limit has to be extracted from the actual experimental linewidths and $g - 2$ data and is not derived here.

6. The Torsion Force as a Hypothesis 6.1 Form of the coupling [H]

The proposed force on a magnetic moment is

$$\vec{F}_{\text{torsion}} = \nabla(\vec{\mu} \cdot \vec{B}_\tau).$$

For a spatially uniform background this force vanishes. A time-dependent $\vec{B}_\tau$ produces a torque ($\vec{\mu} \times \vec{B}_\tau$) and relaxation (Section 5.1), not a net force. A non-zero force therefore needs a gradient of $\vec{B}_\tau$, i.e. localized sources. The "universal force between all massive particles" of the earlier version is **not implied** by the uniform isotropic background and is dropped as a conclusion. It remains a hypothesis that only a non-uniform τ_T field could yield a force.

6.2 Comparison with the established interactions

Interaction	Mediator	Range	Source	Status
Gravitational	graviton (framework label ν_e)	∞	mass	[E] for the interaction; mediator label [T]
Electromagnetic	photon	∞	electric charge	[E]
Weak	$W^{\pm}, Z^0$	$\sim 10^{-18}$ m	weak isospin	[E]
Strong (QCD)	gluons	$\sim 10^{-15}$ m	color charge	[E]
Torsion	τ_T	∞ (hypothetical, stellar origin)	magnetic + light labels	[H], no evidence

No claim is made that a "fifth fundamental force" exists. The proposal is that, *if* it exists, it should show up in the tests listed in Section 6.3.

6.3 Tests and constraints

1. **Null test on the magnetic response.** Absence of any dependence of the magnetic response of nucleons and electrons on local stellar density, location or time bounds B_τ; this is not a test of ZB itself, which has no defined observable for confined quarks.
2. **Spin-coherence / NMR correlation [H].** The earlier-version prediction that NMR relaxation correlates with local stellar density is retained as a *small-effect* test, to be compared with the bounds of Section 5.2. A sidereal or annual modulation would be, in principle, a signature of the preferred frame $\mathcal{S}_0$ in P1, but at the anisotropy level $\sim u/V \approx 10^{-12}$ (Section 4.3e) it lies beyond current NMR sensitivity.
3. **Constituent-quark size (weak, model-dependent: a discrepancy would weaken the identification without excluding it).** The model identifies the nucleon ZB amplitude with $r_Z \approx 0.3$ fm. An independent determination of a constituent-quark size (e.g. from form factors) very different from ≈ 0.3 fm would disfavor the identification.
4. **Precursor timing under P1.** If τ_T were emitted in stellar bursts and could be detected, it would precede light by $\approx d/c$ (Section 4.3b). No detector or dataset is proposed here, and any such test carries the causality caveats of Section 4.4.
5. **Gradient test.** A torsion force requires a non-uniform field (Section 6.1); a null result for anomalous forces on magnetic moments near strong, compact matter concentrations would restrict the hypothesis. The earlier statement that the gyromagnetic ratio is modified near neutron stars is not supported by any computation in this paper and is removed.

7. Conclusions

1. The QCD decomposition of the nucleon spin and $\Delta\Sigma \approx 0.30 \pm 0.05$ are retained as established results. The electron has no quark–gluon structure, and its spin is that of an elementary Dirac fermion.
2. The "trembling" is identified, as a working hypothesis, with Dirac Zitterbewegung of the elementary degrees of freedom. At $p = 0$ (free particle at rest): the electron itself ($\omega_Z = 1.553 \times 10^{21}$ rad/s, $r_Z = 193.1$ fm, period 4.05×10^{-21} s), and, for nucleons, the constituent quarks ($\omega_Z = 9.51 \times 10^{23}$ rad/s, $r_Z = 0.315$ fm, period 6.6×10^{-24} s), with $\omega_Z r_Z = c$ and $mc\, r_Z = \hbar/2$. For a confined quark, ω_Z would be larger by between $\approx 7\%$ ($\Delta p \geq \hbar/2R$: $pc \approx 117$ MeV, $E \approx 334$ MeV) and $\approx 25\%$ ($p \approx \hbar/R$: $pc \approx 235$ MeV, $E \approx 391$ MeV), depending on the confinement estimate; the free-Dirac formula is itself only formal for a confined quark. For the atomic electron the correction is $\approx 3 \times 10^{-5}$. The trembling is set by the mass and $E(p)$, not produced by the torsion particle; any τ_T effect on it is at most a small perturbation, not evaluated here.
3. The torsion particle is retained as a framework hypothesis whose only admissible role is a small perturbation (precession or relaxation), to be bounded by NMR and ESR linewidths and the electron $g - 2$. A precession at the recomputed frequencies of the earlier formula would need $B_\tau \approx 0.38$ T (electron) and ≈ 19 T (proton), which is excluded.
4. The speed $V = 10^9 c$ is a postulate. Its direct consequences are: transit of a proton in 5.6×10^{-33} s, 1 Mpc in 1.03×10^5 s, the Hubble radius in ≈ 14 yr, an undetermined recoil momentum (two options differing by 10^{18}), and a frame anisotropy $\sim u/V \approx 10^{-12}$. It conflicts with Lorentz invariance, causality (frame speeds above $c/10^9 \approx 0.30$ m/s yield reversed ordering), and no-signalling unless a preferred frame and non-controllability of τ_T are assumed. It has no experimental confirmation.

5. Stellar production of τ_T is a postulate (P2) without a verified mechanism. Numerical corrections to the earlier version are in Appendix B.
6. The "fifth force" is a hypothesis; in the uniform isotropic limit its force vanishes. Section 6.3 lists tests that can mostly constrain it.

Appendix A. Model production reaction (framework notation only)

The TauTimp framework writes a stellar production process as

$$^{291}_{115}\mathrm{Mc} + \tau_T \;\rightarrow\; ^{291}_{116}\mathrm{Lv} + \mathrm{op} + \mu + E_m.$$

As written it violates energy conservation ($E_m = 3.59$ eV versus $m_\mu c^2 = 105.66$ MeV [10], a deficit of $\sim 3 \times 10^7$) and lepton-number conservation (a lone muon); "op" is a framework term not defined here; the heaviest moscovium isotope known to the author is ^{290}Mc; ^{291}Mc is therefore treated here as hypothetical. The reaction is therefore reproduced only as notation of the framework and is used nowhere in this paper.
(Check of the deficit: $105.66\ \mathrm{MeV}/3.59\ \mathrm{eV} = 2.94 \times 10^7$.)

Appendix B. Errata to the May 2026 version

The values below were recomputed from the formulas printed in the May 2026 version [14], with $\hbar$ and its inputs ($B = 3.81 \times 10^{14}$ T, e, m_e, m_p, $r_e = 2.82 \times 10^{-15}$ m, $r_p = 0.87 \times 10^{-15}$ m). The inputs B, r_e, r_p are the author's and are not endorsed here.

Quantity	Printed value	Recomputed from the printed formula	Disposition
$\omega_e = \sqrt{5Be\hbar/(m_e r_e^2)}$	1.05×10^{12} s^{-1}	6.67×10^{10} s^{-1}	Withdrawn; not trembling (Section 3)
ω_p (same formula)	5.76×10^{10} s^{-1}	5.04×10^{9} s^{-1}	Withdrawn
$t_p = E_m m_p/(Be\hbar)$	2.4×10^{-8} s	1.49×10^{-7} with $\hbar$ (the printed number matches h)	Withdrawn
$t_e = E_m m_e/(Be\hbar)$	1.3×10^{-11} s	8.1×10^{-11} with $\hbar$	Withdrawn
$v_{\mathrm{surface}} = \omega_e r_e$	$\sim 10^{-3}$ m/s	3×10^{-3} m/s with the printed ω_e	Withdrawn together with ω_e
"Precise agreement" of t_{int}^{-1} with ω	claimed	not reproduced for the proton	Withdrawn

Field strengths at the printed frequencies. With $B_\tau = \hbar\omega/2\mu$: electron at 1.05×10^{12} s^{-1} gives ≈ 6.0 T; proton at 5.76×10^{10} s^{-1} gives ≈ 215 T. Both are excluded by NMR in the Earth's field (5×10^{-5} T). The recomputed frequencies (Section 5.2) give the smaller values 0.38 T and 19 T, scaled linearly in ω.
Additional checks. 1. *Dimensional analysis.* In SI units $B\,e\,\hbar/m$ is an energy (T C J s/kg = J), so E_{spin} and ω are dimensionally consistent. But $E_m m/(Be\hbar)$ has units kg/(T C s) = kg/kg, i.e. it is **dimensionless**, not a time. The "interaction time" t_{int} is therefore removed, together with the identification $\omega \sim t_{\mathrm{int}}^{-1}$. The only time that follows directly from E_m is $\hbar/E_m = 1.83 \times 10^{-16}$ s, which is not used in the arguments. 2. *Non-relativistic energy scale.* With the stated B, $E_{\mathrm{spin}} = Be\hbar/m_e$ exceeds $m_e c^2$ by the factor $B\,e\hbar/(m_e^2 c^2) = B/B_S \approx 8.6 \times 10^4$ for the electron ($B_S = m_e^2 c^2/e\hbar \approx 4.4 \times 10^9$ T). A non-relativistic kinetic-energy balance is then not self-consistent for the electron. 3. *Comparison with ZB.* The printed ω_e is a factor $\sim 1.5 \times 10^9$ below ω_Z (electron), and a rigid rotation at the printed ω_e would have $\omega_e r_e \ll c$, whereas ZB has speed exactly c. The printed numbers therefore describe neither ZB nor anything else derived in this paper.
Removed reference. The entry "Brodsky & Hwang (1998), Phys. Lett. B 553, 223" of the earlier version was removed because its year and volume could not be reconciled and it is not needed for any statement here.

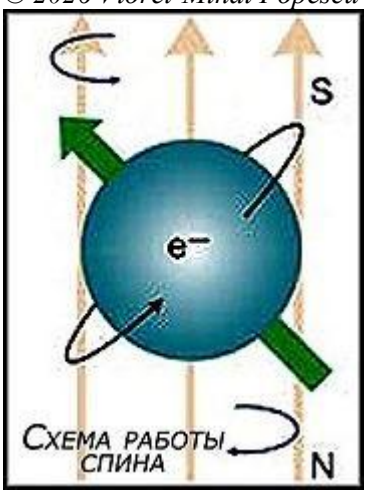
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